GTIIT Spring 2025 Calculus 1M
Workshop 4

March 27, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

1. Let $f(x) = 1/x$ and $g(x) = 1/x^2$.
   (a) Compute $(f \circ g)(x)$
   (b) is $f \circ g$ continuous everywhere?
2. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ such that f is continuous everywhere but in 0 and $|f|$ is continuous everywhere.
3. Use the definition of continuity and the properties of limits to show that the function
$$f(x) = \frac{x^2 + 5x}{2x + 1}$$
is continuous at 2.
4. Find the values of c that make the function f continuous everywhere.
$$f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2. \end{cases}$$
5. Compute the limits:
   (a) $\lim_{x\to\infty} \frac{x^2}{\sqrt{x^4+1}}$
   (b) $\lim_{x\to\infty} \frac{x+3x^2}{4x-1}$
6. Evaluate the limit $\lim_{x\to\infty} \frac{\sin x}{x}$.
7. If $f(x) = 3x^2 - x^3$, find $f'(1)$ and use it to find an equation of the tangent line to the curve $y = 3x^2 - x^3$ at the point $(1, 2)$.
8. Find $f'(a)$

(a) $f(x) = 3x^2 - 4x + 1$

(b) $f(x) = \frac{4}{\sqrt{1-x}}$

9. Give an example of a function that is continuous at $x = 0$ but not differentiable at $x = 0$.

10. If $f(x) = 2x^2 - x^3$, find $f'(x)$, $f''(x)$, $f'''(x)$ and $f^{(4)}(x)$. Graph those functions in geogebra.

11. Prove the following statements:

    (a) The derivative of an even function is an odd function.

    (b) The derivative of an odd function is an even function.

    (c) Give some examples of the previous statements.